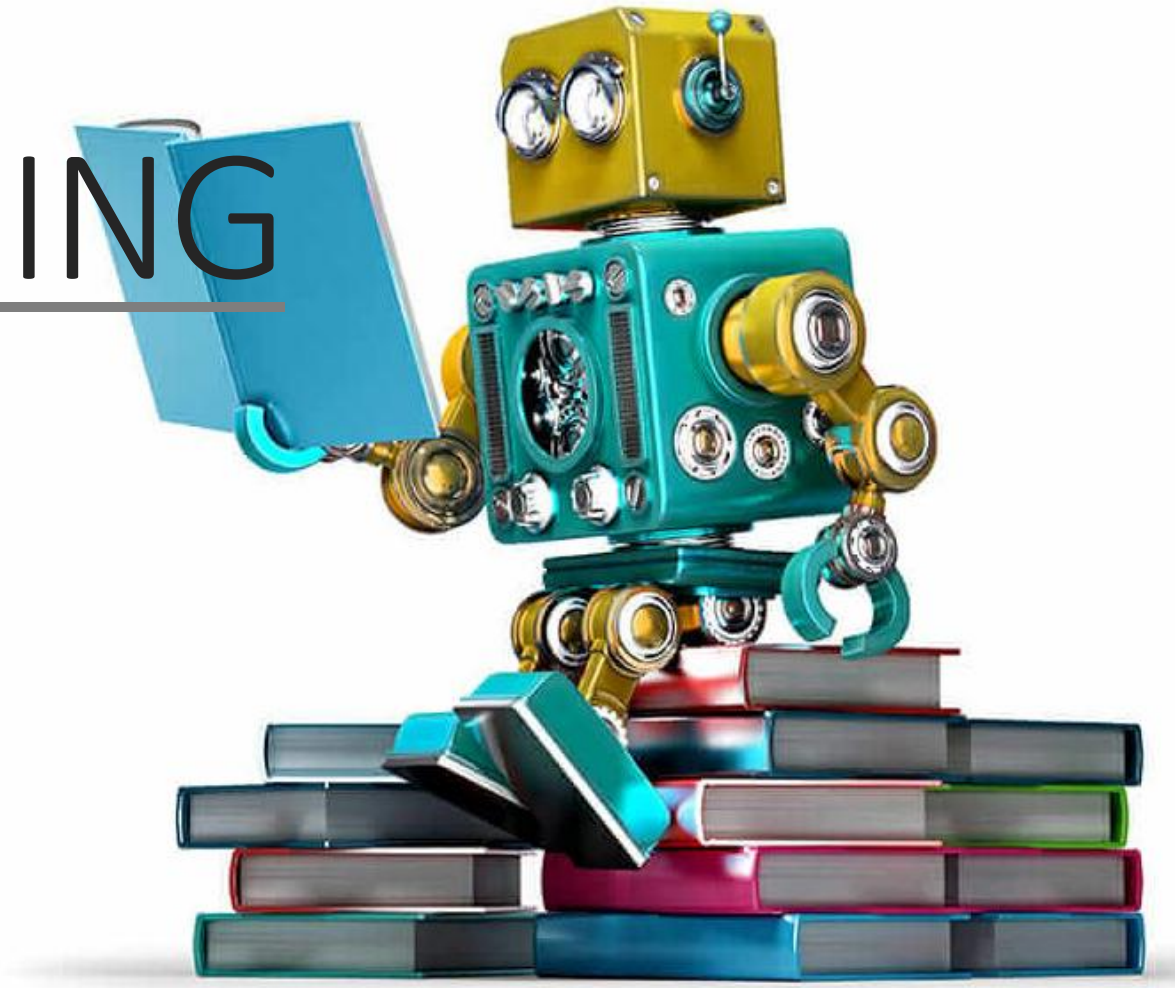
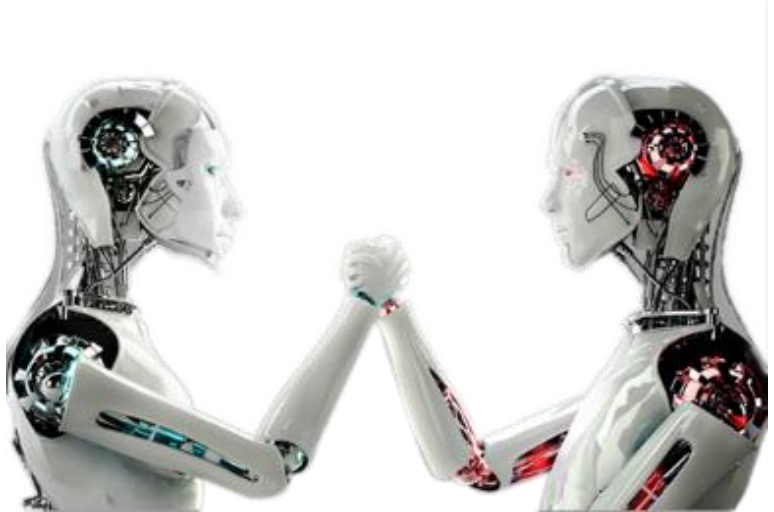
MACHINE LEARNING

LAB1



PART ONE

Lab Introduction



Attendance

- Mandatory
- Be on Time!
- Don't expect labs to end early.
- Let instructor know if you anticipate an absence.

Textbook and Lab Notes

Textbooks:

1 Pattern Recognition and Machine Learning, by Christopher M. Bishop, 2006 Springer

2 Machine Learning in Action, by Peter Harrington, 2012, Manning

3 Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow

Other books:

1 机器学习，周志华

2 Dive in Deep Learning, by Aston zhang, Zachary C.Lipton, Mu Li, and Alexander J.Smola

3 Reinforcement Learning: An Introduction, by Richard S. Sutton

4 the Elements of Statistical Learning Data Mining, Inference, and Prediction

Paper reading:

1 Ghahramani Z. Probabilistic machine learning and artificial intelligence, Nature, 2015

2 Lecun Y, Bengio Y, Hinton G. Deep learning, Nature, 2015

3 Littman M L. Reinforcement learning improves behavior from evaluative feedback, Nature, 2015

Lab Report & Grading Guidelines

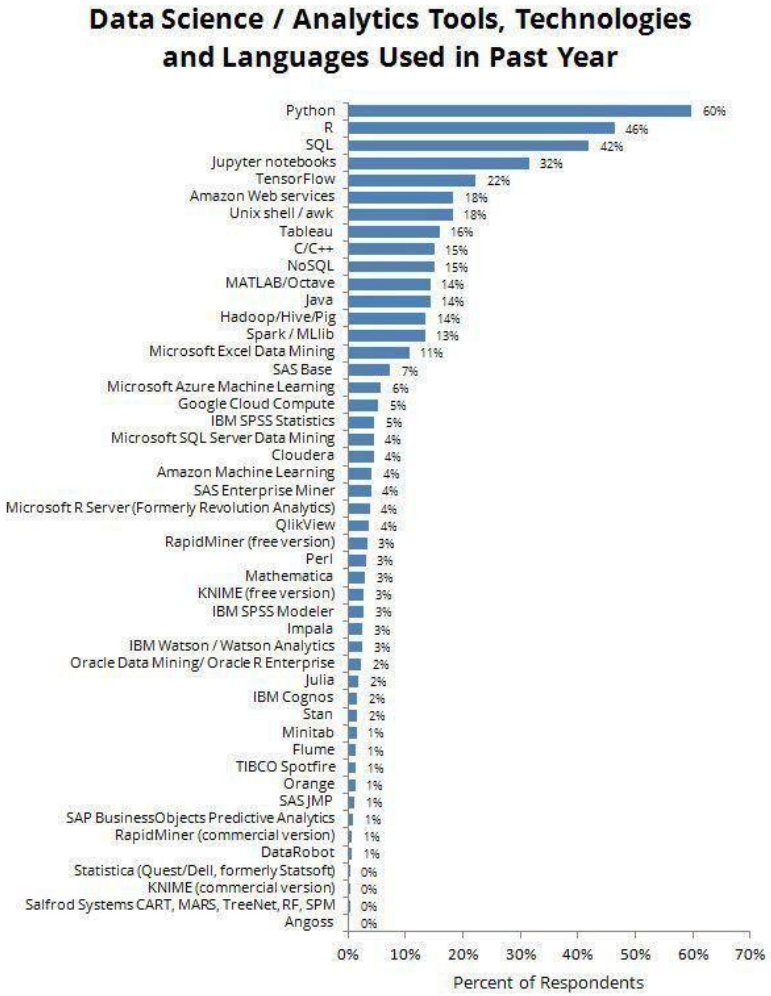
- Lab report
  - ✓ Follow the lab instructions
  - ✓ Testing the algorithms and discuss your observations [25%]
  - ✓ Develop new algorithms (bonus)
  - ✓ Describe your methods
  - ✓ Validating the algorithms and discuss your observations [75%]

we will deduct scores according to this rule, but we will not strictly follow it all the time. In some cases, we may decide to detract more or less points based on the perceived gravity of the mistakes.

Websites

- **Lab Tutotial web sites**
  - *http://scikit-learn.org/stable/tutorial/basic/tutorial.html*
  - *https://pythonprogramming.net/python-fundamental-tutorials/*
  - *https://pythonprogramming.net/data-analysis-tutorials/*
  - *https://pythonprogramming.net/robotics-tutorials/*
  - http://prml.github.io/
  - https://scikit-learn.org/dev/user_guide.html

Which Programming language is popular for ML?



Data are from the Kaggle 2017 The State of Data Science and Machine Learning study. You can learn more about the study and download the data here: https://www.kaggle.com/surveys/2017.
Respondents were asked to indicate for work, which data science/analytics tools, technologies, and languages they used in the past year. A total of 10153 respondents answered the question.

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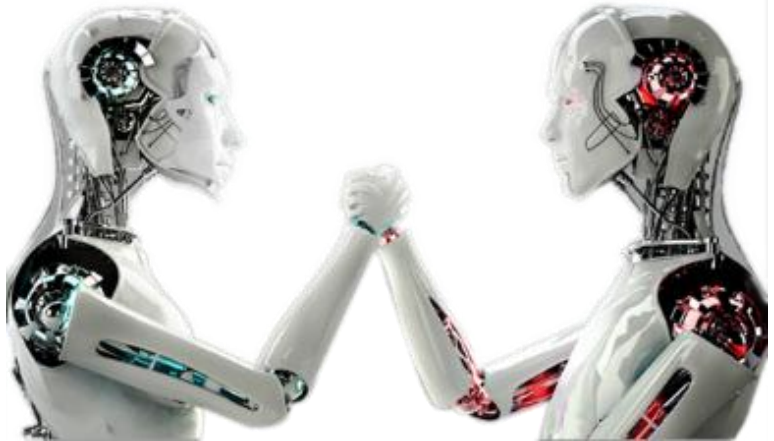
software

- software
  - Python
    - numpy — 数值数组和矩阵类型和它们的基本运算
    - scipy — 使用NumPy来做高等数学、信号处理、优化、统计和许多其它科学任务
    - pandas — 基于NumPy的一种工具，为了解决数据分析任务而创建的
    - matplotlib — 用来生成绘图,直方图,功率谱,条形图,错误图,散点图等
    - scikit-learn — 机器学习库,基本功能主要被分为六大部分:分类、回归、聚类、数据降维、模型选择和数据与处理
  - Open AI Gym — 用于开发和比较RL 算法的工具包
  - Deep Learning framework
    - Pytorch — Facebook
    - torch — NYU/Facebook
    - Tensorflow — Google
    - Caffe
    - MXNet — 李沐，亚马逊官方框架
    - PaddlePaddle — 百度

PART TWO

Prepare the development environmentIntroduction



Prepare the development env intro

1. Install python + vscode

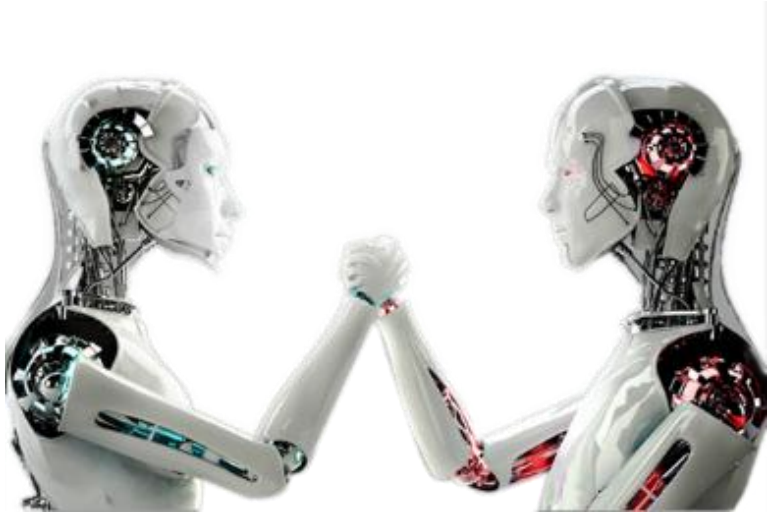
- See lab meterial.

2. Install essential libraries for Machine Learning in Python

- **Numpy,** a package with support for mathematical operations on multidimensional data—was the most imported package, used in nearly three-quarters of machine learning and data science projects.（https://numpy.org/）
- **scipy,** a package for scientific computation.（https://www.scipy.org/）
- **pandas,** a very popular library,it can easily fetch data from different sources like SQL databases, CSV, Excel, JSON files and manipulate the data to perform operations on it. （https://pandas.pydata.org/）
- **matplotlib,** a visualization library.（https://matplotlib.org/）
- **scikit-learn ,**Scikit-learn is one the most popular ML libraries. It supports many supervised and unsupervised learning algorithms. Examples include linear and logistic regressions, decision trees, clustering, k-means and so on.For a novice in ML, Scikit-learn is a more-than-sufficient tool to work with, until you start implementing more complex algorithms.（https://scikit-learn.org/stable/index.html）

PART THREE

Lab Task



Lab Task

Install and configure your development environment on your own computer.

Create new Python environment. Run HelloWorld.py and standardplot.py in this environment.